Do Now · retrieval

Data structures: OCR exam practice

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

When an item is popped from a stack, is the oldest or the most recently pushed item removed first?

Q2

Queue: which item is removed first?

Q3

What do left/right columns contain in an array-based BST?

Q4

What four actions does OCR 1.4.2(c) require?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Stack wording — CODE:
stack → LIFO
push → add to top
pop → remove from top — OCR questions often test both the operation and the consequence.

Core idea 2

Queue wording — CODE:
queue → FIFO
enqueue → rear
dequeue → front — State the ordering difference clearly rather than saying only 'they work differently'.

Core idea 3

Linked-list wording — CODE:
start pointer → first item
next pointer → index of next item
null → end — OCR June 2023 explicitly tests how the second item is accessed using pointer values.

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

BST table wording — CODE:
Left | Data | Right
pointer | value | pointer — OCR examiners reported that candidates often incorrectly put data rather than pointer indexes in left/right columns.

Vocabulary · use the correct technical terms

LIFO — Last In, First Out.

FIFO — First In, First Out.

pointer — A value identifying another array location.

record — Related fields describing one item.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Turn understanding into a 2-mark comparison

Follow the worked method

1. Name stack rule — CODE:
Stack removes the most recently added item first.

2. Link the two — CODE:
Therefore the same input order produces different removal orders.

3. Use exam terms — CODE:
LIFO versus FIFO

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Improve the answer
CODE:
An array and a list store multiple values, but ...

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Compare structures

Answer the four short questions comparing stack/queue, array/list, tuple/list, and linked-list pointer access.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Create / traverse / add / remove
For linked list, stack, queue, graph, BST and hash table, state how one add or remove operation works.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

In OCR's June 2024 BST table, the left and right columns contain pointer/index values rather than copies of town names.

Use key vocabulary from memory

1. Explain LIFO in your own words.

2. Explain FIFO in your own words.

3. Explain pointer in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.